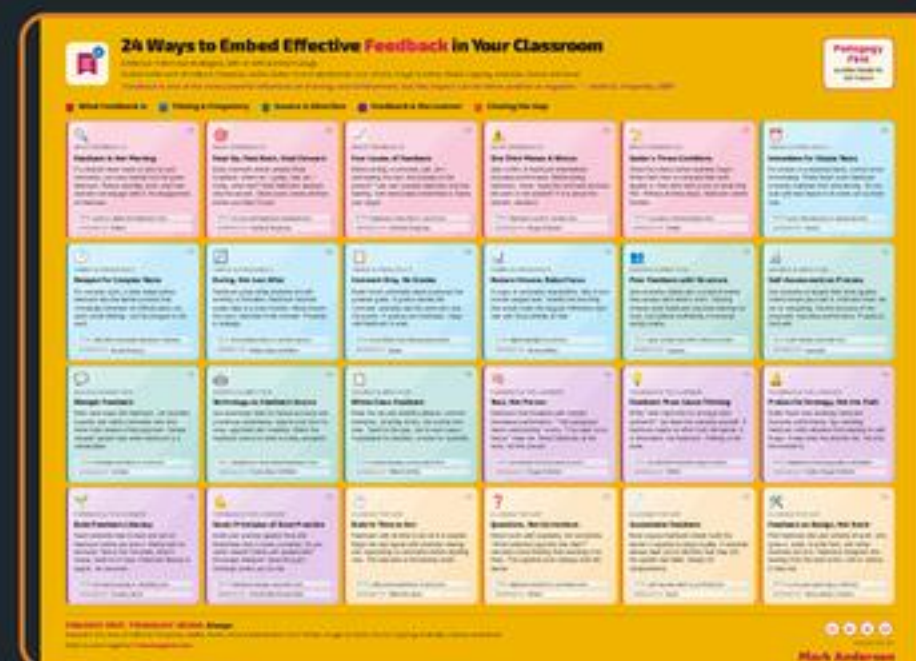
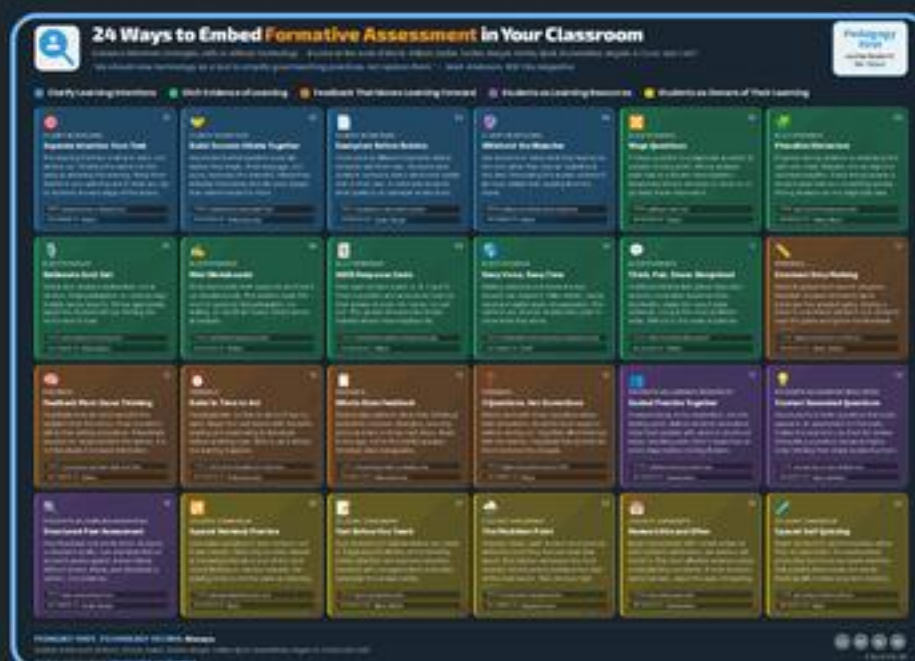


Pedagogy First. Technology Second.

Six guides exploring the science of learning, and the thinking behind one hundred and forty four strategies to try, with or without technology.





24 Ways to Embed Retrieval Practice in Your Classroom

Evidence-informed strategies, with or without technology · Rooted in the work of Bjork, Roediger & Karpicke, Agarwal, Dunlosky, Cepeda, Kornell, Rohrer, Ebbinghaus, Kapur and Metcalfe

"The act of retrieving knowledge is itself a learning event, not merely a test of what has been learned." — Roediger & Karpicke, 2006

Pedagogy First

24 STRATEGIES TO TRY TODAY

The Testing Effect Spacing & Interleaving Desirable Difficulties Retrieval Formats Metacognition & Calibration

01

THE TESTING EFFECT

Retrieval Beats Re-reading

Replace re-reading with retrieval. Instead of reviewing notes, ask students to write down everything they remember, then check. The testing is the learning, not the preparation for it.

TECH low-stakes quiz or flashcard tool

INFORMED BY Roediger & Karpicke

02

THE TESTING EFFECT

Low-Stakes Starter Quizzes

Five questions per lesson on prior content; nothing in the gradebook. Agarwal found routine quizzing reduces test anxiety rather than causing it. The quiz is not the assessment. It is the learning.

TECH quiz or response tool

INFORMED BY Roediger & Karpicke; Agarwal et al.

03

THE TESTING EFFECT

Brain Dump

A blank page, two minutes, no notes: write everything you remember. The imperfect attempt is the point. What students cannot recall shows precisely where teaching needs to return.

TECH shared doc or whiteboard tool

INFORMED BY Roediger; Karpicke; Agarwal

04

THE TESTING EFFECT

Test Before You Teach

Quiz students before teaching. Struggling to retrieve primes the brain; students who fail before instruction remember the answer more reliably once taught. Failure is productive.

TECH pre-lesson quiz tool

INFORMED BY Bjork; Kapur

05

THE TESTING EFFECT

Closed-Book Reconstruction

Students reconstruct key ideas from memory, then open notes and correct in a different colour. Karpicke and Blunt found this outperforms re-reading. The corrections are the data.

TECH document tool; colour annotation

INFORMED BY Karpicke & Blunt

06

SPACING & INTERLEAVING

Distribute Practice Over Time

Spread retrieval across days and weeks rather than cramming it into one session. Build review points into your medium-term plan. The calendar is part of the pedagogy.

TECH spaced repetition tool

INFORMED BY Cepeda et al.; Ebbinghaus

07

SPACING & INTERLEAVING

Yesterday, Last Week, Last Month

Three questions per lesson: one from yesterday, one from last week, one from last month. These intervals are a starting point; complex topics may need revisiting more frequently.

TECH quiz tool; lesson slide template

INFORMED BY Rosenshine; Bjork

08

SPACING & INTERLEAVING

Interleave Topics, Don't Block

Kornell and Bjork found mixing topics produces stronger retention than finishing one before moving to the next. Blocked practice feels organised; it produces weaker retention. Discriminating is the mechanism.

TECH shuffled quiz or random question tool

INFORMED BY Kornell & Bjork; Rohrer

09

SPACING & INTERLEAVING

The Retrieval Schedule

Dunlosky's review ranked re-reading among the least effective strategies. Teach students the evidence: study, return after two days, then a week, then a month. Many students do the opposite.

TECH spaced repetition or revision planner

INFORMED BY Dunlosky et al.

10

SPACING & INTERLEAVING

Cumulative Review

Plan retrieval into your medium-term planning, not just lessons. Bring back earlier content in starters and assessments. When it never reappears, students conclude it no longer matters.

TECH question bank or quiz builder

INFORMED BY Bjork; Carey

11

DESIRABLE DIFFICULTIES

Embrace the Struggle

Bjork coined desirable difficulties for conditions that slow apparent learning but improve long-term retention. Effortful retrieval is the clearest example. Resist the urge to make it easier.

TECH any retrieval tool; hints disabled

INFORMED BY R. Bjork

12

DESIRABLE DIFFICULTIES

Remove the Scaffolding

Retrieval with notes open is reading, not retrieval. Fade support gradually: notes first, then headings only, then nothing. The transition to unaided retrieval is where consolidation happens.

TECH quiz tool; timed or closed-book mode

INFORMED BY R. Bjork; Kornell

13

DESIRABLE DIFFICULTIES

Vary the Retrieval Context

Retrieve the same content in different formats and settings, decontextualising knowledge so it is not tied to one context or format. Harder for students; that is precisely the point.

TECH varied formats; multi-modal tools

INFORMED BY Godden & Baddeley; Bjork

14

DESIRABLE DIFFICULTIES

The Generation Effect

Slamecka and Graf found generating an answer, even a wrong one, before being told it produces stronger retention than passive exposure. The effort of generation drives encoding, not the accuracy.

TECH response tool; pre-teach quiz

INFORMED BY Slamecka & Graf; McDaniel

15

DESIRABLE DIFFICULTIES

Forget Before You Retrieve

Do not retrieve immediately after teaching. Wait a day, then quiz. Wait a week, then quiz again. The gap between learning and retrieval is where consolidation happens.

TECH spaced quiz tool; lesson timer

INFORMED BY R. Bjork; Roediger

16

RETRIEVAL FORMATS

Retrieval Beyond Facts

Factual retrieval builds the schema; higher-order retrieval tests whether students can use it. Secure the foundations first, then push beyond them. Retrieval is not a pub quiz.

TECH quiz tool with varied question types

INFORMED BY Agarwal

17

RETRIEVAL FORMATS

Concept Mapping from Memory

Karpicke and Blunt found concept mapping from memory outperformed re-reading for retention. Not just what students know but how ideas connect. Relational knowledge beats isolated facts.

TECH mind-mapping or diagram tool

INFORMED BY Karpicke & Blunt

18

RETRIEVAL FORMATS

Retrieval Through Teaching

The learning happens during preparation, not delivery. Preparing to teach demands complete, organised retrieval. Organising knowledge to explain it is where consolidation occurs.

TECH video recording or presentation tool

INFORMED BY Nestojko et al.; Fiorella

19

RETRIEVAL FORMATS

Cued Recall

Tulving and Thomson showed retrieval depends on available cues. A heading, a keyword, labels removed: each triggers retrieval without doing the work. Builds confidence before unaided retrieval.

TECH cloze quiz or gapped handout tool

INFORMED BY Tulving & Thomson

20

METACOGNITION & CALIBRATION

Teach the Illusion of Knowing

Show students their own illusion of knowing. Ask them to rate confidence before revealing the answer. When they were certain and wrong, that is the precise moment to intervene.

TECH confidence-rated quiz tool

INFORMED BY Koriati & Bjork; Dunlosky

21

METACOGNITION & CALIBRATION

Provide Instant Feedback

Metcalfe shows instant feedback is essential for confident errors. For general retrieval, allow a processing gap before revealing answers. Speed matters most when confidence is highest.

TECH quiz tool with instant answer reveal

INFORMED BY Metcalfe

22

METACOGNITION & CALIBRATION

Confidence Ratings

Dunlosky and Metcalfe showed students are most confident about things most recently read, not things they actually know. Rate confidence after each attempt, then reveal the answer. That gap is the target.

TECH quiz tool with confidence rating

INFORMED BY Dunlosky & Metcalfe

23

METACOGNITION & CALIBRATION

Retrieval Journals

Zimmerman showed students who monitor their performance make better revision decisions. Log what came back, what did not, and what needs revisiting. It is a map of what to do next.

TECH learning journal or digital notebook

INFORMED BY Dunlosky et al.; Zimmerman

24

METACOGNITION & CALIBRATION

Study Strategy Audit

Dunlosky's review rated ten strategies: re-reading scored low; practice testing scored highest. Ask students how they revise, then show them the data. That conversation is the intervention.

TECH survey or discussion tool

INFORMED BY Dunlosky et al.

PEDAGOGY FIRST, TECHNOLOGY SECOND. Always.

Rooted in the work of Bjork, Roediger & Karpicke, Agarwal, Dunlosky, Cepeda, Kornell, Rohrer, Ebbinghaus, Kapur, Metcalfe, Nestojko, Fiorella & Mayer, Slamecka & Graf and Godden & Baddeley

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CREATED BY

Mark Anderson

Six resources, grounded in the **cognitive science** of **teaching and learning**.

Each guide is built around what the research actually says, not what happens to be trending. The aim was never to show off what technology can do. It was to start with the teaching, work out what genuinely helps learning, and only then ask where a tool might serve that purpose. Sometimes it does, and sometimes the best thing in the room has no battery in it at all.

What looks like a single page of strategies sits on top of decades of classroom practice, leadership, research reading and applied work. That is the point of a distillation: the simplicity on the surface is earned by the depth underneath. Use these ideas, adapt them, argue with them, and make them your own.

| *Pedagogy first. Technology second. Always.*



24 Ways to Embed Formative Assessment in Your Classroom

Evidence informed strategies, with or without technology · Rooted in the work of Black, William, Butler, Sadler, Berger, Hattie, Bjork, Rosenshine, Angelo & Cross and CAST
"We should view technology as a tool to amplify good teaching practices, not replace them." — Mark Anderson, RISE Edu Magazine

Pedagogy First

24 STRATEGIES TO TRY TODAY

- Clarify Learning Intentions
- Elicit Evidence of Learning
- Feedback That Moves Learning Forward
- Students as Learning Resources
- Students as Owners of Their Learning

01

CLARIFY INTENTIONS

Separate Intention from Task

The learning intention is what to learn, not what to do. Finishing the task is not the same as achieving the learning. Keep them distinct in your planning and in what you say to students at every stage of the lesson.

TECH presentation or display tool

INFORMED BY William

02

CLARIFY INTENTIONS

Build Success Criteria Together

Ask students what excellent looks like before they begin. Their language, not yours, becomes the standard. Criteria they articulate themselves land far more deeply than criteria handed to them.

TECH collaborative document tool

INFORMED BY William & Leahy

03

CLARIFY INTENTIONS

Exemplars Before Rubrics

Show work at different standards before students start their own. Students spot quality in someone else's work more readily than in their own. A rubric tells students what quality is; an exemplar shows them.

TECH visualiser or document camera

INFORMED BY Sadler, Berger

04

CLARIFY INTENTIONS

Withhold the Objective

Ask students to name what they learned at the end rather than copy an objective at the start. Generating the answer embeds it far more reliably than copying from the board.

TECH polling tool; whole class response

INFORMED BY William

05

ELICIT EVIDENCE

Hinge Questions

A hinge question is a diagnostic question at a lesson turning point. Wrong answers each map to a known misconception. Responses tell you whether to move on or go back. Never improvise it.

TECH polling or quiz tool

INFORMED BY William

06

ELICIT EVIDENCE

Plausible Distractors

Engineer wrong answers as carefully as the right one. Each distractor should map to a real misconception. Every wrong answer a student gives tells you something precise. Wrong answers are the diagnostic data.

TECH quiz tool; full response view

INFORMED BY William, Black

07

ELICIT EVIDENCE

Deliberate Cold Call

Select who answers deliberately, not at random. Track participation so nobody stays invisible across lessons. Where appropriate, target the students whose thinking you most need to hear.

TECH participation tracking tool

INFORMED BY William, Black

08

ELICIT EVIDENCE

Mini Whiteboards

All students write their response and hold it up simultaneously. The teacher reads the room in seconds: full participation, no waiting, no dominant voices. Works across all subjects.

TECH multimodal response tools

INFORMED BY William

09

ELICIT EVIDENCE

ABCD Response Cards

Give each student a card: A, B, C and D. Pose a question and all students hold up their answer at once. No hands, no opt-out. The spread of responses shows instantly where misconceptions lie.

TECH multimodal audience response tools

INFORMED BY William

10

ELICIT EVIDENCE

Every Voice, Every Time

Eliciting evidence only works if every student can respond. Offer written, verbal, visual and digital means of expression. The method you choose decides who gets to show what they know.

TECH multimodal and anonymous response tools

INFORMED BY CAST

11

ELICIT EVIDENCE

Think, Pair, Share: Disciplined

Individual thinking first; paired discussion second; nominated response third. Nomination makes the report-back evidence, not just the most confident voice. Without it, the share is optional.

TECH timer for paired discussion

INFORMED BY William

12

FEEDBACK

Comment Only Marking

Remove grades from work in progress. Research showed comments alone produced the greatest gains. Adding a score to a comment cancels it out: students read the grade and ignore the feedback.

TECH digital annotation or LMS tool

INFORMED BY Butler, William

13

FEEDBACK

Feedback Must Cause Thinking

Feedback must be more work for the recipient than the donor. Pose questions rather than writing corrections. If feedback requires no response from the learner, it is not feedback; it is merely information.

TECH annotation tool with reply function

INFORMED BY William

14

FEEDBACK

Build In Time to Act

Feedback with no time to act on it has no value. Begin the next lesson with students reading and responding to feedback before anything new. Time to act is where the learning happens.

TECH LMS; share feedback in advance

INFORMED BY William & Leahy

15

FEEDBACK

Whole Class Feedback

Record class patterns rather than individual comments: common strengths, recurring errors and two or three next steps. Teach to the gap, not to the individual paper. Workload stays manageable.

TECH shared document or display tool

INFORMED BY William & Hattie

16

FEEDBACK

3 Questions, Not Corrections

Return work with three questions rather than corrections. Students must respond before moving on. Cognitive effort belongs with the learner. Feedback that corrects for them removes the struggle.

TECH digital annotation tool or LMS

INFORMED BY William

17

STUDENTS AS LEARNING RESOURCES

Guided Practice Together

Independence is the destination, not the starting point. Before students work alone, have them practise with peers in structured steps, checking each other's responses at every stage before moving forward.

TECH collaborative document tool

INFORMED BY Rosenshine

18

STUDENTS AS LEARNING RESOURCES

Student Generated Questions

Ask students to write questions that could appear in an assessment on this topic. Collate the best and use them for review. Generating questions demands higher order thinking than simply answering them.

TECH shared doc or quiz builder tool

INFORMED BY Black & William

19

STUDENTS AS LEARNING RESOURCES

Structured Peer Assessment

Peer feedback only works when students understand quality. Use exemplars first so students assess against shared criteria. Without shared criteria, peer feedback is opinion, not evidence.

TECH peer assessment tool

INFORMED BY Sadler, Berger

20

STUDENT OWNERSHIP

Spaced Retrieval Practice

Low stakes quizzing on prior content, not today's lesson. Returning to older material at increasing intervals is one of the most robust findings in memory research. Re-reading notes is not the same as retrieving.

TECH spaced repetition quiz tool

INFORMED BY Bjork

21

STUDENT OWNERSHIP

Test Before You Teach

Quiz students on material before you teach it. Triggering the feeling of not knowing primes attention and improves retention. Students who struggle before instruction remember the answer better.

TECH quiz tool; lesson start

INFORMED BY Bjork, William

22

STUDENT OWNERSHIP

The Muddiest Point

At lesson close, each student anonymously writes the one thing they are least clear about. The teacher addresses the most common muddy points publicly at the start of the next lesson. Two minutes; high impact.

TECH anonymous response tool

INFORMED BY Angelo & Cross

23

STUDENT OWNERSHIP

Review Little and Often

Begin every lesson with a brief review of prior content: yesterday's, last week's, last month's. The most effective teachers check understanding constantly. If most students cannot answer, adjust the pace of teaching.

TECH low stakes quiz tool

INFORMED BY Rosenshine

24

STUDENT OWNERSHIP

Spaced Self Quizzing

Teach students to quiz themselves rather than re-read notes. Re-reading feels productive but produces weak retention. Self quizzing feels harder but works. Retrieval effort drives long term memory.

TECH quizzing or flashcard tool

INFORMED BY Bjork

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James Donaldson · Assistant Principal, Cardiff and Vale College



24 Ways to Embed Effective Feedback in Your Classroom

Evidence-informed strategies, with or without technology
Rooted in the work of Hattie & Timperley, Sadler, Butler, Nicol & Macfarlane-Dick, William, Kluger & DeNisi, Shute, Topping, Andrade, Carless and Boud

"Feedback is one of the most powerful influences on learning and achievement, but this impact can be either positive or negative." – Hattie & Timperley, 2007

Pedagogy First
24 STRATEGIES TO TRY TODAY

- What Feedback Is
- Timing & Frequency
- Source & Direction
- Feedback & the Learner
- Closing the Gap

01

WHAT FEEDBACK IS

Feedback Is Not Marking

If a student never reads or acts on your comments, you have marked but not given feedback. Before returning work, plan how students will engage with it. No engagement, no feedback.

TECH audio or digital ink feedback tool

INFORMED BY William

02

WHAT FEEDBACK IS

Feed Up, Feed Back, Feed Forward

Every comment should answer three questions: where am I going, how am I doing, what next? Most feedback answers only the second. Check yours covers all three before you hand it back.

TECH structured feedback template tool

INFORMED BY Hattie & Timperley

03

WHAT FEEDBACK IS

Four Levels of Feedback

Before writing a comment, ask: am I addressing the task, the process or the person? Task and process feedback improve learning. Self-level praise undermines it. Name your target.

TECH feedback checklist or rubric tool

INFORMED BY Hattie & Timperley

04

WHAT FEEDBACK IS

One Third Makes It Worse

Over a third of feedback interventions decrease performance. Before giving feedback, check: does this comment address the work or the student? If it is about the student, rewrite it.

TECH feedback audit or review tool

INFORMED BY Kluger & DeNisi

05

WHAT FEEDBACK IS

Sadler's Three Conditions

Share the criteria before students begin. Show them how to compare their work against it. Give them time to act on what they find. Without all three steps, feedback cannot function.

TECH success criteria display tool

INFORMED BY Sadler

06

TIMING & FREQUENCY

Immediate for Simple Tasks

For simple or procedural tasks, correct errors immediately. Shute found quick feedback prevents mistakes from embedding. Do not wait until next lesson to fix what can be fixed now.

TECH auto-marked quiz or response tool

INFORMED BY Shute

07

TIMING & FREQUENCY

Delayed for Complex Tasks

For complex work, a short delay before feedback lets the learner process first. Immediate correction on difficult tasks can short-circuit thinking. Let the struggle do its work.

TECH LMS with scheduled feedback release

INFORMED BY Shute; Kulhavy

08

TIMING & FREQUENCY

During, Not Just After

Feedback given while students are still working is formative. Feedback returned weeks later is a post-mortem. Move around the room, intervene in the moment. Proximity is strategy.

TECH live collaboration or monitoring tool

INFORMED BY William; Black & William

09

TIMING & FREQUENCY

Comment Only, No Grades

Butler found comments alone produced the greatest gains. A grade cancels the comment: students see the score and skip the words. If grades are mandated, delay until feedback is used.

TECH annotation tool; hide grades option

INFORMED BY Butler

10

TIMING & FREQUENCY

Reduce Volume, Raise Focus

A page of comments overwhelms. One or two precise targets land. Identify the one thing that would make the biggest difference right now and focus entirely on that.

TECH digital highlight or pin tool

INFORMED BY Shute; William

11

SOURCE & DIRECTION

Peer Feedback with Structure

Give students criteria and a protocol before they assess each other's work. Topping showed peer feedback improves learning for both, but without scaffolding it becomes empty praise.

TECH peer review tool with criteria prompts

INFORMED BY Topping

12

SOURCE & DIRECTION

Self-Assessment as Process

Ask students to assess their work against criteria before you mark it. Andrade found the act of comparing, not the accuracy of the judgment, improves performance. Process is the point.

TECH self-review checklist tool

INFORMED BY Andrade

13

SOURCE & DIRECTION

Dialogic Feedback

Build reply loops into feedback. Let students question and clarify comments over time rather than receive a final judgment. Carless showed uptake rises when feedback is a conversation.

TECH threaded comment or chat tool

INFORMED BY Carless

14

SOURCE & DIRECTION

Technology as Feedback Source

Use automated tools for factual accuracy and procedural correctness; reserve your time for voice, argument and creativity. Match the feedback source to what is being assessed.

TECH adaptive or AI-assisted feedback tool

INFORMED BY Shute; Black & William

15

SOURCE & DIRECTION

Whole Class Feedback

Read the set and identify patterns: common strengths, recurring errors, one priority next step. Teach to the gap, not to each paper. Sustainable for teachers; precise for students.

TECH shared display or document tool

INFORMED BY William; Hattie

16

FEEDBACK & THE LEARNER

Task, Not Person

Feedback that threatens self-concept decreases performance. "This paragraph needs restructuring" works. "You need to try harder" does not. Direct feedback at the work, not the person.

TECH annotation tool focused on work

INFORMED BY Kluger & DeNisi

17

FEEDBACK & THE LEARNER

Feedback Must Cause Thinking

Write "why might this be stronger with evidence?" not insert the evidence yourself. If feedback needs no effort from the learner, it is information, not feedback. Thinking is the point.

TECH annotation tool with reply function

INFORMED BY William

18

FEEDBACK & THE LEARNER

Praise the Strategy, Not the Trait

Butler found task-involving feedback improves performance. Ego-involving feedback shifts attention from learning to self-image. Praise what the student did, not who the student is.

TECH feedback phrasing guide or template

INFORMED BY Butler; Kluger & DeNisi

19

FEEDBACK & THE LEARNER

Build Feedback Literacy

Teach students how to read and act on feedback before you give it. Model with an exemplar: here is the comment, what it means, what to do next. Feedback literacy is taught, not assumed.

TECH worked example or modelling tool

INFORMED BY Carless; Boud

20

FEEDBACK & THE LEARNER

Seven Principles of Good Practice

Audit your practice against Nicol and Macfarlane-Dick's seven principles. Do you clarify criteria? Enable self-assessment? Encourage dialogue? Close the gap? Redesign where you do not.

TECH feedback design checklist tool

INFORMED BY Nicol & Macfarlane-Dick

21

CLOSING THE GAP

Build In Time to Act

Feedback with no time to act on it is wasted. Begin the next lesson with students reading and responding to comments before anything new. The response is the learning event.

TECH LMS; share feedback in advance

INFORMED BY William & Leahy

22

CLOSING THE GAP

Questions, Not Corrections

Return work with questions, not corrections. "What evidence supports this claim?" demands more thinking than rewriting it for them. The cognitive work belongs with the learner.

TECH digital annotation or comment tool

INFORMED BY William

23

CLOSING THE GAP

Sustainable Feedback

Boud argues feedback should build the learner's capacity to judge quality. If students always need you to tell them how they did, the system has failed. Design for independence.

TECH self-assessment or portfolio tool

INFORMED BY Boud

24

CLOSING THE GAP

Feedback as Design, Not Event

Plan feedback into your scheme of work: who gives it, when, in what form, and when students act on it. Feedback designed into learning from the start works. Left to chance, it does not.

TECH curriculum planning or LMS tool

INFORMED BY Boud; Sadler; Carless

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24 Ways to Embed Effective Questioning & Discussion in Your Classroom

Evidence-informed strategies, with or without technology · Rooted in cognitive and educational research

Rooted in the work of Anderson & Krathwohl, Dillon, Hattie & Zierer, Rowe, Tobin, Lyman, William, Good & Brophy, Alexander, Mercer, Michaels, O'Connor & Resnick, and King

"Classroom talk is the most pervasive but arguably the least examined tool in any teacher's repertoire." — Robin Alexander

Pedagogy First

24 STRATEGIES TO TRY TODAY

The Quality of Questions

Wait Time & Thinking Space

Participation & Equity

Dialogic Teaching & Discussion

Student-Generated Questions



01

THE QUALITY OF QUESTIONS

Questions That Make Thinking Visible

Anderson and Krathwohl map six levels: remembering, understanding, applying, analysing, evaluating, creating. These are not a hierarchy of worth. Recall is foundational, not inferior.

TECH question bank or lesson planning tool

INFORMED BY Anderson & Krathwohl



02

THE QUALITY OF QUESTIONS

Surface, Deep and Transfer

Hattie and Zierer distinguish surface (what), deep (why) and transfer questions. Each serves a different stage. Reaching transfer before surface is secure sets students up to fail.

TECH discussion prompt or slide display tool

INFORMED BY Hattie & Zierer



03

THE QUALITY OF QUESTIONS

Open Questions Demand More

Dillon found genuine open questions, where the teacher does not know the answer, produce the most extended student reasoning. Not every question needs a predetermined destination.

TECH shared whiteboard or discussion board

INFORMED BY Dillon



04

THE QUALITY OF QUESTIONS

Plan Questions Before the Lesson

Improvised questions default to recall, not because recall is wrong but because planning stops there. Anderson and Krathwohl works as a planning tool: write questions at every level.

TECH lesson planning or slide preparation tool

INFORMED BY Anderson & Krathwohl



05

THE QUALITY OF QUESTIONS

Alternatives to Questions

Dillon found statements and silence elicit richer responses than questions. Try: Tell me more. I had not considered that. Not every probe needs to end with a question mark.

TECH discussion or annotation tool

INFORMED BY Dillon



06

WAIT TIME & THINKING SPACE

Wait Time 1: After You Ask

Rowe found most teachers wait under one second before speaking. Extending to three seconds increases the length and complexity of responses. Silence is thinking, not a gap to fill.

TECH timer or countdown display tool

INFORMED BY Rowe



07

WAIT TIME & THINKING SPACE

Wait Time 2: After a Student Speaks

Rowe identified a second wait time: the pause after a student responds. Three seconds prompts elaboration and self-correction. Silence after an answer signals that more is expected.

TECH live polling or response capture tool

INFORMED BY Rowe



08

WAIT TIME & THINKING SPACE

Write Before You Speak

Ask students to write before any verbal response is taken. Writing prevents the quickest from foreclosing others' thinking. The pen democratises the room before the voice takes over.

TECH shared doc, whiteboard or response tool

INFORMED BY Tobin; Rowe



09

WAIT TIME & THINKING SPACE

Think-Pair-Share

Lyman's think-pair-share gives every student thinking time before public exposure. Pairs test ideas with a peer first. The result is more considered and more substantive discussion.

TECH partner chat tool or response system

INFORMED BY Lyman



10

PARTICIPATION & EQUITY

Hands Down

William shows hand-raising distorts who answers. A small group volunteers repeatedly; the rest disengage. Remove hands and direct to named students. Every student should be asked.

TECH random name picker or classlist tool

INFORMED BY William



11

PARTICIPATION & EQUITY

Cold Call Without Threat

Good and Brophy showed differential questioning reinforces attainment gaps. Direct questions are equitable only when the climate supports it. Norms: no wrong attempts, only thinking.

TECH random selector or seating plan tool

INFORMED BY Good & Brophy; William



12

PARTICIPATION & EQUITY

Track Who You Ask

Good and Brophy found teachers direct harder questions to higher-attaining students. Track participation across lessons. Patterns invisible in the moment become visible on paper.

TECH participation tracker or tally tool

INFORMED BY Good & Brophy



13

PARTICIPATION & EQUITY

Pose, Pause, Pounce, Bounce

William's PPPB: pose the question, pause, direct to one student, then bounce the answer to another. The bounce keeps all students alert and spreads thinking across the whole room.

TECH display timer; random name tool

INFORMED BY William



14

DIALOGIC TEACHING & DISCUSSION

Five Principles of Dialogic Talk

Alexander identifies five qualities of dialogic talk: collective, reciprocal, supportive, cumulative and purposeful. Most talk meets two or three of these. Audit and start with the gaps.

TECH lesson observation or reflection tool

INFORMED BY Alexander



15

DIALOGIC TEACHING & DISCUSSION

Break the IRE Pattern

IRE (Initiation, Response, Evaluation): ask, answer, judge. Alexander shows this pattern limits thinking. Replace evaluation with a follow-up question. What makes you say that?

TECH video reflection or lesson capture tool

INFORMED BY Alexander; Mercer



16

DIALOGIC TEACHING & DISCUSSION

Exploratory Talk

Mercer distinguishes disputational, cumulative and exploratory talk. Only exploratory talk, where students challenge ideas with evidence and build on each thinking, improves outcomes.

TECH collaborative doc or discussion tool

INFORMED BY Mercer



17

DIALOGIC TEACHING & DISCUSSION

Ground Rules for Talk

Mercer found teaching norms of exploratory talk transforms discussion. Norms co-constructed with the class outperform those displayed or modelled. Make the ground rules together.

TECH shared doc or display tool

INFORMED BY Mercer



18

DIALOGIC TEACHING & DISCUSSION

Accountable Talk

Michaels, O'Connor and Resnick hold students to accurate knowledge and rigorous reasoning. Teach moves: I agree because..., The evidence is..., Say more.... Language shapes thinking.

TECH discussion sentence starter display

INFORMED BY Michaels, O'Connor & Resnick



19

DIALOGIC TEACHING & DISCUSSION

Build on Student Responses

Alexander's EEF programme showed gains when teachers extended rather than evaluated. Return an answer: Aisha says X. Does anyone want to add or challenge? Conductor, not judge.

TECH live display or response capture tool

INFORMED BY Alexander



20

STUDENT-GENERATED QUESTIONS

Teach Question Stems

King found students trained to generate questions using stems produce richer discussion than those answering only teacher questions. Generating questions demands deeper processing.

TECH question stem cards or display tool

INFORMED BY King



21

STUDENT-GENERATED QUESTIONS

Reciprocal Peer Questioning

King's reciprocal questioning has students generate and pose questions to each other, with neither knowing the answer. This produces genuine enquiry rather than rehearsal. Aim to think.

TECH pair chat or response tool

INFORMED BY King



22

STUDENT-GENERATED QUESTIONS

Questions as Evidence of Understanding

A student's question reveals the depth of their understanding. Ask for one written question before leaving. Why does X cause Y? shows far more than What is X?. Questions are windows.

TECH exit ticket or response tool

INFORMED BY King; William



23

STUDENT-GENERATED QUESTIONS

Interthinking Through Questions

Mercer's interthinking describes shared cognition built through language. Student questions sustain joint enquiry. The goal is thinking in common, not performance for the teacher.

TECH collaborative discussion or writing tool

INFORMED BY Mercer



24

STUDENT-GENERATED QUESTIONS

Questions That Drive Enquiry

King found connecting questions to prior knowledge produces the most powerful learning. Ask students to write what they know and want to know. Their questions become the entry point.

TECH shared doc, padlet or survey tool

INFORMED BY King

PEDAGOGY FIRST, TECHNOLOGY SECOND. Always.

Rooted in the work of Anderson & Krathwohl, Dillon, Hattie & Zierer, Rowe, Tobin, Lyman, William, Good & Brophy, Alexander, Mercer, Michaels, O'Connor & Resnick, and King

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CREATED BY

Mark Anderson



24 Ways to Embed Effective Explanations & Modelling in Your Classroom

Evidence-informed strategies, with or without technology · Rooted in cognitive and educational research
Rooted in the work of Rosenshine, Sweller, Mayer, Paivio, Willingham, Atkinson, and the EEF Digital Technology Guidance (2019)
“To improve learning, technology must be used in a way that is informed by effective pedagogy.” — EEF, 2019

Pedagogy First
24 STRATEGIES TO TRY TODAY

- Structuring Explanations
- Managing Cognitive Load
- Multimedia & Dual Coding
- Modelling & Worked Examples

01

STRUCTURING EXPLANATIONS

Small Steps, Always

Rosenshine: present new material in small steps with practice after each. Use animations to chunk information and reveal content gradually. Let students process before continuing.

TECH slide animation; progressive reveal

INFORMED BY Rosenshine

02

STRUCTURING EXPLANATIONS

Concrete Before Abstract

Willingham: anchor new ideas to concrete examples first. Use images, video or objects to make the abstract visible before naming it. Move to the abstract once the concrete is secure.

TECH visualiser; image or video clip

INFORMED BY Willingham

03

STRUCTURING EXPLANATIONS

Connect to Prior Knowledge

Rosenshine: connect to prior knowledge before introducing anything new. Use a retrieval starter or concept map on screen to activate what students know. Link before you launch.

TECH concept map or retrieval tool

INFORMED BY Rosenshine

04

STRUCTURING EXPLANATIONS

Clarity Over Novelty

The EEF found clarity matters more than delivery method. A diagram under a visualiser can outperform a polished video. Choose the tool that makes the idea clearest, not the newest.

TECH any display tool used with purpose

INFORMED BY EEF

05

STRUCTURING EXPLANATIONS

Check, Don't Assume

Listening is not learning. Use live polls or mini whiteboards to check understanding at every step. Rosenshine: reteach before moving on. The explanation is the start, not the finish.

TECH live polling or mini whiteboard tool

INFORMED BY Rosenshine

06

STRUCTURING EXPLANATIONS

Supplement, Never Substitute

The EEF: technology works best when it supplements teaching, not replaces it. A screencast works alongside classroom explanation, not instead of it. Supplement, never substitute.

TECH screencast; LMS for replay access

INFORMED BY EEF

07

MANAGING COGNITIVE LOAD

Four Slots, No More

Sweller: working memory holds about four items. Every screen element competes for those slots. Strip slides to essentials. Press B to blank the screen when you want attention on you.

TECH B/W key; clean slide design

INFORMED BY Sweller

08

MANAGING COGNITIVE LOAD

Avoid Split Attention

Sweller's split attention: separate text and diagrams force learners to search between them. Integrate labels directly into the image. Use annotation tools to build the diagram live.

TECH annotation tool; integrated diagrams

INFORMED BY Sweller

09

MANAGING COGNITIVE LOAD

Say It or Show It, Not Both

Sweller's redundancy effect: narrating text on screen forces double processing. Say it or show it, not both. Use the B or W key to blank the slide while you talk through the idea.

TECH B/W key in PowerPoint or Keynote

INFORMED BY Sweller

10

MANAGING COGNITIVE LOAD

Remove the Clutter

Mayer's coherence: remove every word, image and sound that does not support learning. Strip clip art from slides. Cut background music. Every element on screen must earn its place.

TECH clean slide template; minimal design

INFORMED BY Mayer

11

MANAGING COGNITIVE LOAD

Signal What Matters

Mayer's signalling: bold, arrows and highlights direct attention to what matters. Use annotation tools to signal live during the explanation. Without it, learners search alone.

TECH annotation; slide highlight features

INFORMED BY Mayer

12

MANAGING COGNITIVE LOAD

Chunk and Reveal

Mayer's segmenting: break content into chunks. Use slide animations to reveal one idea per click, not auto-play. Let working memory process before the next chunk arrives.

TECH slide animation; click to reveal

INFORMED BY Mayer

13

MULTIMEDIA & DUAL CODING

Words and Images Together

Paivio's dual coding: the brain processes words and images through separate channels. Design every slide with both: a key visual paired with minimal text. Two routes to the same idea.

TECH dual coded slide design

INFORMED BY Paivio

14

MULTIMEDIA & DUAL CODING

Narrate, Don't Label

Mayer's modality: people learn better from images with narration than images with text. Record a narrated screencast over a diagram. Let the voice do what text would overload.

TECH screen recording with voiceover

INFORMED BY Mayer

15

MULTIMEDIA & DUAL CODING

Keep Words Near Images

Mayer's spatial contiguity: place words near the images they describe. Labels sit beside the diagram, not in a separate list. Use slide design to keep related information together.

TECH integrated slide layout

INFORMED BY Mayer

16

MULTIMEDIA & DUAL CODING

Progressive Reveal

Use animations to reveal content one step at a time. One idea per click, building the full picture gradually. Progressive reveal aligns the display to the pace of your explanation.

TECH slide animation; appear on click

INFORMED BY Mayer; Sweller

17

MULTIMEDIA & DUAL CODING

Short Screencasts

Record short screencasts with narration over visuals. Mayer: images plus voice outperforms text-heavy slides. Keep each video to one idea. Brevity beats completeness every time.

TECH screen recording tool; Loom or similar

INFORMED BY Mayer

18

MULTIMEDIA & DUAL CODING

Pre-Train the Vocabulary

Mayer's pre-training: teach key terms before the main explanation. A vocabulary slide or pre-lesson video builds familiarity first. Learners who know the words can focus on the ideas.

TECH vocabulary slide; pre-lesson video

INFORMED BY Mayer

19

MODELLING & WORKED EXAMPLES

Think Aloud

Rosenshine: model the process, not the product. Use a visualiser or screen share to think aloud live. Narrate your decisions as you work. Make invisible expertise visible to all.

TECH visualiser; screen share with narration

INFORMED BY Rosenshine

20

MODELLING & WORKED EXAMPLES

The Visualiser as Stage

A visualiser turns any desk into a modelling stage. Show handwriting, annotate text, build diagrams under the camera. The EEF found this among the most promising classroom technology.

TECH visualiser or document camera

INFORMED BY EEF; Rosenshine

21

MODELLING & WORKED EXAMPLES

Worked Examples on Replay

Sweller and Cooper: students learn more from worked examples than equivalent problems. Record a screencast of the example so students can pause and study the method at their pace.

TECH screen recording; LMS for sharing

INFORMED BY Sweller & Cooper

22

MODELLING & WORKED EXAMPLES

Alternate Examples and Practice

Atkinson: alternate worked examples with practice. Display an example on screen, then set a parallel task. Example, try. Example, try. Interleaving builds understanding faster.

TECH display and task alternating on screen

INFORMED BY Atkinson; Sweller

23

MODELLING & WORKED EXAMPLES

Fade the Scaffolds

Rosenshine: begin with full support, then fade. Show a complete example, then a partial one with gaps, then independent practice. Use fillable templates to structure the fading.

TECH fillable template or shared doc

INFORMED BY Rosenshine; Sweller

24

MODELLING & WORKED EXAMPLES

Simulations Beyond Physical Limits

The EEF found simulations let students manipulate variables and see consequences beyond physical limits. Use to extend your explanation, not replace it. Technology serves pedagogy.

TECH simulation tool; interactive model

INFORMED BY EEF

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24 Ways to Embed Metacognition & Self-Regulation in Your Classroom

Evidence-informed strategies, with or without technology · Rooted in cognitive and educational research
Rooted in the work of Zimmerman, Flavell, Winne, Butler, and Willingham

“As children should learn about metacognition and how best to learn, they should also be taught how to use technology to support their learning.” — Anderson & Lewis, *The EdTech Playbook*, 2025

Pedagogy First
24 STRATEGIES TO TRY TODAY

- Forethought & Planning
- Performance & Monitoring
- Self-Reflection & Evaluation
- Strategy Knowledge & Use
- Digital Cognition

01

FORETHOUGHT & PLANNING

Set the Goal First

Zimmerman: planning begins with a specific, proximal goal, not a vague aim. Have students state what a good answer to this question looks like before they start writing it.

TECH shared success criteria on screen

INFORMED BY Zimmerman

02

FORETHOUGHT & PLANNING

Knowledge Comes First

Willingham: you cannot plan or monitor a task you know nothing about. Teach the content first; metacognition works on knowledge, never instead of it. Build the base, then regulate.

TECH retrieval starter to check the base

INFORMED BY Willingham

03

FORETHOUGHT & PLANNING

Analyse the Task First

Zimmerman: strong learners break a task into its demands before starting. Ask students what this question is really asking and which prior topics it draws on before they answer.

TECH annotate the question on a board

INFORMED BY Zimmerman

04

FORETHOUGHT & PLANNING

Pick a Strategy on Purpose

Flavell: planning means selecting an approach, not defaulting to one. Have students name which method they will use for this problem and why, before the first line goes down.

TECH strategy menu in a shared doc

INFORMED BY Flavell

05

FORETHOUGHT & PLANNING

Build Realistic Confidence

Zimmerman: self-efficacy drives effort, but it must be earned by past success, not pep talks. Point students to a recent task they mastered to set a credible expectation for this one.

TECH portfolio of prior work to revisit

INFORMED BY Zimmerman

06

PERFORMANCE & MONITORING

Check, Don't Just Feel

Winne: a novice's sense of understanding is unreliable. Pair every self-check with an external test, a quiz or worked answer, so monitoring rests on evidence rather than confidence.

TECH self-marking quiz with feedback

INFORMED BY Winne

07

PERFORMANCE & MONITORING

Scaffold Focus, Then Fade

Zimmerman: monitoring needs a low-distraction environment first. Use classroom.cloud to scaffold focus while students are learning, then withdraw it as they manage attention themselves.

TECH classroom.cloud

INFORMED BY Zimmerman

08

PERFORMANCE & MONITORING

Build in Stop Points

Winne: monitoring fades without prompts. Insert planned pauses where students ask whether the approach is still working, before sunk effort makes a wrong method hard to abandon.

TECH timed checkpoint prompts

INFORMED BY Winne

09

PERFORMANCE & MONITORING

Ask Yourself As You Go

Flavell: monitoring means questioning your understanding mid-task. Teach students to pause and ask “do I actually understand this?”, so gaps surface in time to fix them.

TECH self-check prompts mid-task

INFORMED BY Flavell

10

PERFORMANCE & MONITORING

Make Thinking Visible

Flavell: regulation you cannot see, you cannot coach. Have students leave a brief trace of their reasoning as they work, so you and they can inspect the process, not just the product.

TECH shared doc showing working e.g. OneNote

INFORMED BY Flavell

11

SELF-REFLECTION & EVALUATION

Judge Against a Standard

Zimmerman: self-judgement needs a reference, not a hunch. Give students the mark scheme or an exemplar to compare their work against, so evaluation is anchored to a real standard.

TECH exemplar shown beside own work

INFORMED BY Zimmerman

12

SELF-REFLECTION & EVALUATION

Credit the Method, Not the Gift

Zimmerman: how students explain a result shapes the next attempt. Steer attribution toward the method used, which they can change, rather than fixed ability, which they cannot.

TECH reflection prompt after results

INFORMED BY Zimmerman

13

SELF-REFLECTION & EVALUATION

Turn Reflection Into a Plan

Butler: feedback only regulates learning if it changes the next action. Have students convert each reflection into one concrete adjustment they will make on the very next task.

TECH action log carried forward

INFORMED BY Butler

14

SELF-REFLECTION & EVALUATION

Compare Predicted to Actual

Winne: calibration improves when students see the gap between expected and real scores. Have them predict a mark, then compare, so monitoring is corrected by evidence over time.

TECH prediction logged before marking

INFORMED BY Winne

15

SELF-REFLECTION & EVALUATION

Reflection Feeds Teaching

Butler: self-regulation and instruction are a loop. Use what students notice about their own errors to decide what you reteach, so their reflection shapes the next lesson.

TECH exit poll of common errors

INFORMED BY Butler

16

STRATEGY KNOWLEDGE & USE

Know the Strategy Itself

Flavell: students cannot deploy a strategy they have never been taught. Teach retrieval, spacing or self-explanation explicitly as named methods before expecting independent use.

TECH strategy reference sheet

INFORMED BY Flavell

17

STRATEGY KNOWLEDGE & USE

Know When to Use Each

Winne: conditional knowledge, knowing which strategy suits which task, is what separates skilled learners. Teach not just the method but the cues that signal when it is the right one.

TECH worked cases matched to methods

INFORMED BY Winne

18

STRATEGY KNOWLEDGE & USE

Fade the Scaffold

Zimmerman: support must be withdrawn for self-regulation to develop. Move from modelled to guided to independent use, removing prompts as students show they no longer need them.

TECH prompts gradually removed

INFORMED BY Zimmerman

19

STRATEGY KNOWLEDGE & USE

Keep What Works, Drop the Rest

Winne: learners refine an approach by checking whether it worked. Have students trial a study method, judge it against results, and keep only what the evidence supports.

TECH method log with outcomes

INFORMED BY Winne

20

STRATEGY KNOWLEDGE & USE

Tie Strategies to Content

Willingham: strategies do not float free of subjects. Teach each one inside the content it serves, so students learn to self-regulate this topic, not metacognition in the abstract.

TECH subject-specific strategy banks

INFORMED BY Willingham

21

DIGITAL COGNITION

Schedule Your Own Spacing

Regulating when you study is a planning act. Teach students to set their own calendar reminders to revisit topics over time, owning the spacing rather than waiting to be told.

TECH calendar reminders for review

INFORMED BY Zimmerman

22

DIGITAL COGNITION

Make Your Own Questions

Writing a good question forces you to judge what matters. Have students build low-stakes quizzes and flashcards from their own notes, so the tool supports retrieval they have authored.

TECH self-built quizzes; flashcard app

INFORMED BY Winne

23

DIGITAL COGNITION

Regulate the Device Too

Self-regulation now includes the screen. Teach students to use focus modes and silence alerts as a deliberate choice, regulating the tool so it serves attention instead of stealing it.

TECH focus mode; app limits

INFORMED BY Zimmerman

24

DIGITAL COGNITION

Partner, Not Substitute

Willingham: memory is the residue of thought. Offload the thinking to AI and nothing is learned. Have students use it to check and challenge their reasoning, not to do it for them.

TECH AI as a critic of own work

INFORMED BY Willingham

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